# What Is a System of Equations?

An introduction to understanding two unknowns and two conditions

## Start with a question

Suppose you buy 4 tickets to a school event. Adult tickets cost $8 and student tickets cost $5. You spend exactly $26.

How many adult tickets and how many student tickets did you buy?

## Why one equation is not enough

Let a represent the number of adult tickets and s represent the number of student tickets.

$$a + s = 4$$

This tells us the total number of tickets. But there are several possibilities: 0 adults and 4 students, 1 adult and 3 students, 2 adults and 2 students, and so on.

We also know the total cost:

$$8a + 5s = 26$$

This second equation adds another condition. We need a pair of numbers that makes both equations true at the same time.

## The big idea

A system of equations is a set of two or more equations considered together. A solution is a value, or a set of values, that satisfies every equation in the system.

For our tickets, a = 2 and s = 2. Check: 2 + 2 = 4 tickets, and 8(2) + 5(2) = $26. Both conditions work.

## Why do we use systems?

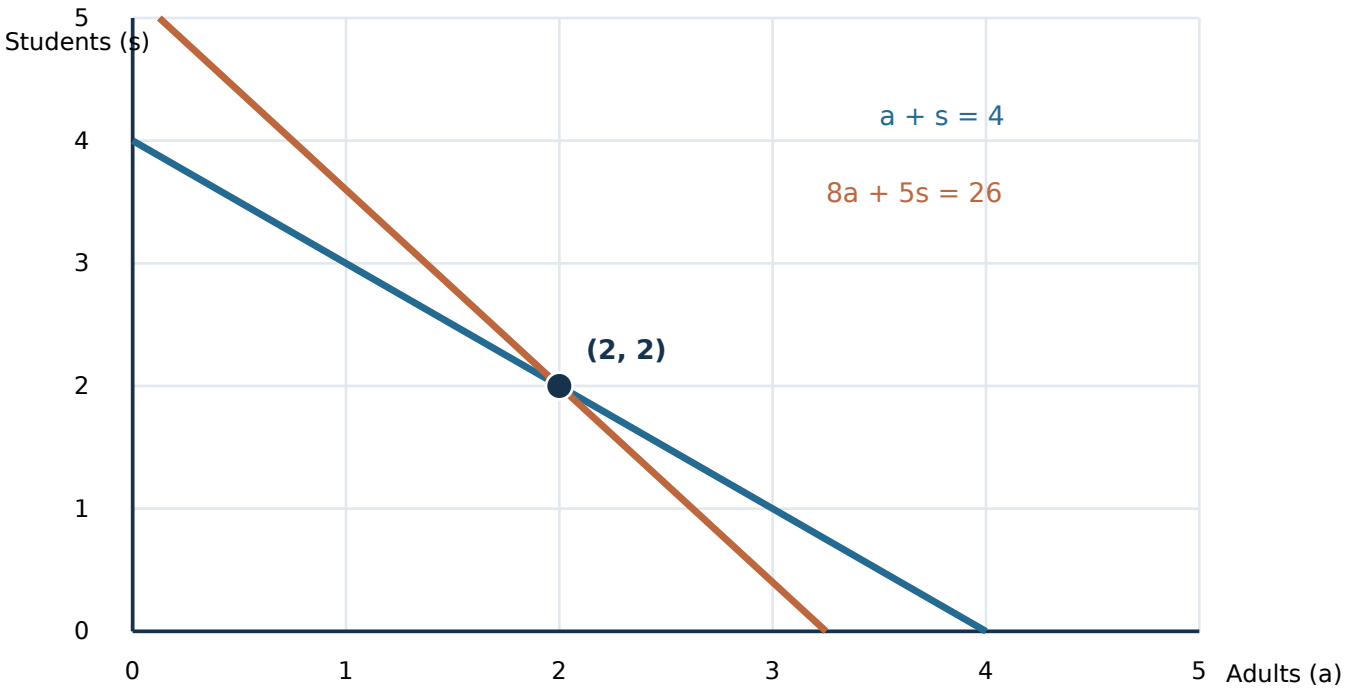
Real situations often give us several facts about several unknowns. Systems let us combine those facts to find an answer that works for all of them. Examples include comparing phone plans, finding prices of two items, and figuring out when two travelers meet.

# Seeing a System on a Graph

Each equation describes many possible pairs. The solution is where their graphs meet.

Ticket equations, rewritten for graphing:

**s = 4 − a and s = 5.2 − 1.6a**




The intersection (2, 2) means 2 adult tickets and 2 student tickets. It is on both lines, so it satisfies both equations.

## An important distinction

One equation with two unknowns often has many solutions. A system asks us to find the solution(s) shared by all equations. Two lines can cross once, never cross (parallel lines), or lie on top of one another (the same line). So a system can have one solution, no solution, or infinitely many solutions.

## Talk about it before learning a procedure

1. Why does the equation a + s = 4 not tell us exactly which tickets were bought?

2. What additional information does the cost equation give us?

3. Why must we check an answer in both equations?

4. What does the intersection of the two lines mean in the story?

Takeaway: A system combines multiple conditions. Solving it means finding values that make every condition true.